

3N+1 PROBLEM 4

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The $3N+1$ sequence is generated using the following rules: If n is even, the next number is $n / 2$. If n is odd, the next number is $3 \times n + 1$. The sequence continues until it reaches 1. The cycle length of a number is the total number of terms in the sequence, including the starting number and 1.

Given two integers i and j , determine the maximum cycle length among all integers between i and j (inclusive).

If $i > j$, swap them before processing.

Input Format

A single line containing two space-separated integers:

i j

Constraints

$1 \leq i, j \leq 1,000,000$

Output Format

Print three integers:

i j maximum_cycle_length

using the original values of i and j .

Sample Input 0

1 10

Sample Output 0

1 10 20

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Python 3



```

1 i , j=map(int,input().split())
2 if(i>j):
3     i,j=j,i
4 arrcount=[]
5 d=0
6 for k in range(i,j+1):
7     count=1
8     n=k

```

```
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9      while n!=1:
10          count+=1
11          if n%2==0:
12              n=n//2
13          else:
14              n=(3*n)+1
15          arrcount.append(count)
16      print(i,j,max(arrcount),end=" ")
17
```

Line: 17 Col: 1



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Test against custom input

Run Code

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